

Hello there!

CS 374A Review MT2

Electric Boogaloo

ACM @ UIUC

~~November 8, 2025~~

April 11th, 2026 :



Disclaimers and Logistics

- This review session is being recorded. Recordings and slides will be distributed on EdStem after the end.
- **Agenda:** We'll quickly review all topics likely to be covered, then go through a practice exam, then review individual topics by request.
- Please let us know if we're going too fast/slow, not speaking loud enough/speaking too loud, etc.
- If you have a question anytime during the review session, please ask! Someone else almost surely has a similar question.
- We'll provide a feedback form at the end of the session.

Recursion

- **Definition:** Reducing the problem to a smaller instance of itself, where eventually we can terminate in a base case.
 - Think: If we have a problem of size n , we want to continuously reduce to a problem smaller than n .
 - Example: Binary search, Karatsuba

Template

```
1: procedure AMAZINGRECURSIVEALGO( $n$ )
2:   if  $n ==$  [some base case] then
3:     return [value]
4:   else
5:     return AmazingRecursiveAlgo( $n/2$ )
```

- Similar to **induction**!

Recursion: Runtime Analysis

- **General Form:**

$$T(n) = \underbrace{r}_{\substack{\text{\# of subproblems}}} \cdot \overbrace{T\left(\frac{n}{c}\right)}^{\text{work at each subproblem}} + \underbrace{f(n)}_{\text{work at current level}}$$

- Describes how the amount of work changes between each level of recursion.
- We can solve for a **time complexity** that describes the scaling behavior of the algorithm at hand.

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- **Master Theorem**

Master Theorem

Decreasing: $r \cdot f(n/c) = \kappa \cdot f(n)$ where $\kappa < 1 \implies T(n) = O(f(n))$

Equal: $r \cdot f(n/c) = \kappa \cdot f(n)$ where $\kappa = 1 \implies T(n) = O(f(n) \cdot \log_c n)$

Increasing: $r \cdot f(n/c) = \kappa \cdot f(n)$ where $\kappa > 1 \implies T(n) = O(n^{\log_c r})$

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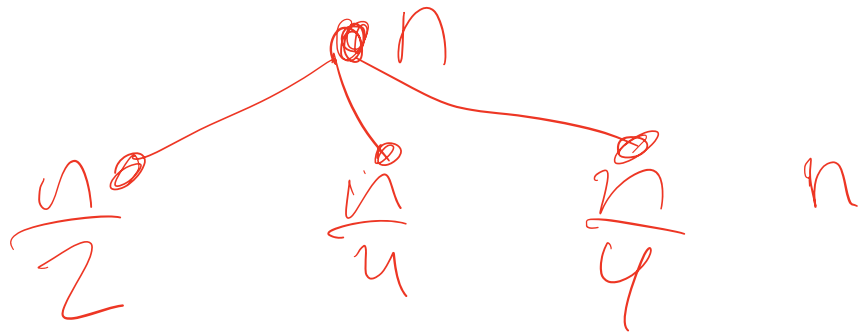
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- **Intuition:** If each level contains more work than the level below it, then the root level will dominate. If each level contains the same amount of work, then we have $\log_c n$ levels with $f(n)$ work. If each level contains less work than the work below it, then the leaf nodes will dominate.

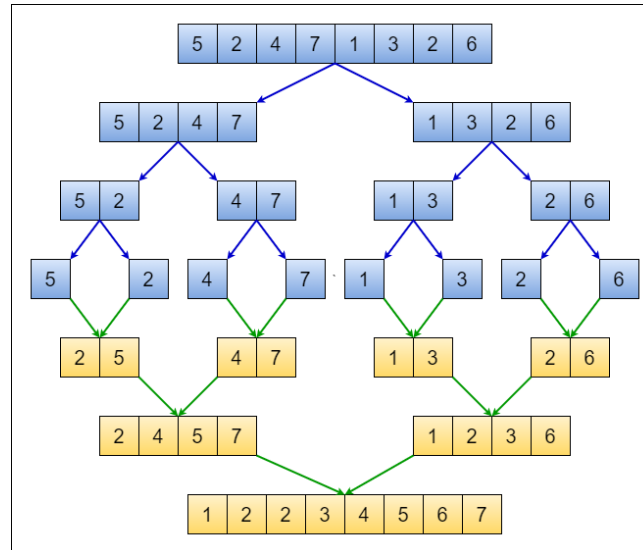
$$T(n) = 2T\left(\frac{n}{4}\right) + \underbrace{T\left(\frac{n}{2}\right)}_n + O(n)$$



$$O(n \log n)$$

Divide and Conquer Algos: Merge Sort

- **Purpose:** Sort an arbitrary array.
- **Time Complexity:** $O(n \log n)$
- **Intuition:** Three phases: (a) split the array in half, (b) sort each side, (c) merge the sorted halves by repeatedly comparing smallest elements on each side not yet inserted.

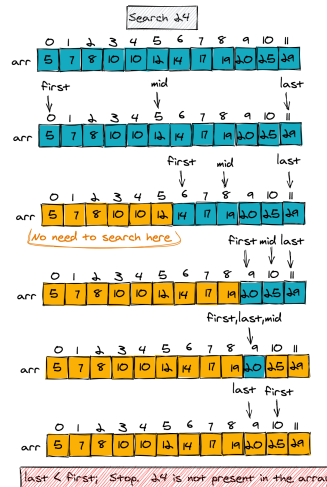


Divide and Conquer Algos: MomSelect

- **Purpose:** Get the r^{th} smallest element in an arbitrary array.
- **Time Complexity:** $O(n)$
- **Intuition:** Pick a pivot P with a value P_V and rearrange the array such that all the elements that are less than P_V are to the left of P and all the elements that are greater than P_V are to the right of P . If the length of the array of elements that are less than P_V is greater than r , then we know that the r^{th} smallest element is to the left of P and we recurse on the left subarray. Otherwise, we know that the r^{th} smallest element is to the right of P and we recurse on the right subarray.

Divide and Conquer Algos: Binary Search

- **Purpose:** Find the existence of an element in a sorted array
- **Time Complexity:** $O(\log n)$
- **Intuition:** Say we are trying to find the value k . Pick the middle element M in the array. If $k > M$, the element must be to the right of M and we recurse on the right. Otherwise, we recurse on the left.



Dynamic Programming

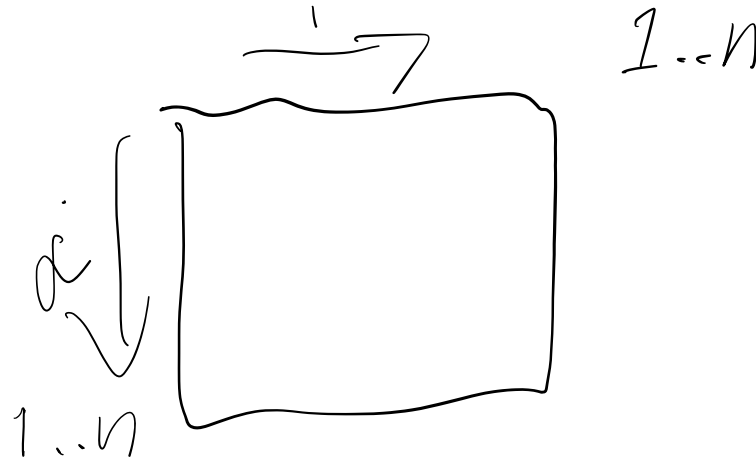
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- For a DP solution, we need:
 1. English Description
 2. Recurrence
 3. Memoization Structure
 4. Solution Location
 5. Evaluation Order
 6. Runtime



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- For a DP solution, we need:
 1. English Description
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 6. Runtime
- **How to solve a DP:**
 - Identify how we can take advantage of a recursive call on a smaller subset of the input space.
 - Identity base cases
 - Identity recurrences (they should cover all possible cases at each step)

$[1, 1, 10, 3, 4]$



Dynamic Programming: Longest Increasing Subsequence

$A[1..n]$

1. **English Description:** $LIS(i, j)$ = the length of the longest increasing subsequence in $A[i..n]$, such that every element in the subsequence is $> A[j]$. Here, $j = 0$ can be used as a sentinel with $A[0] = -\infty$.

2. **Recurrence:**

$$LIS(i, j) = \begin{cases} 0 & \text{if } i > n, \\ LIS(i + 1, j) & \text{if } A[i] \leq A[j], \\ \max(LIS(i + 1, j), 1 + LIS(i + 1, j)) & \text{if } A[i] > A[j]. \end{cases}$$

3. **Memoization Structure:** A 2D array $LIS[0..n, 0..n]$, storing all subproblem solutions $LIS(i, j)$.
4. **Solution Location:** The length of the LIS of the entire array is found at $LIS(1, 0)$, starting from the first element with the sentinel.
5. **Evaluation Order:** Fill the table in decreasing order of i .
6. **Runtime:** The table has $O(n^2)$ entries, each computed in $O(1)$, so the total runtime is $O(n^2)$.

Graphs

- **Definition:** A set of vertices V connected by a set of edges E . Individual edges are notated as (u, v) , where $u, v \in V$.
 - They are usually represented as **adjacency lists** or **adjacency matrices**

$$u \rightarrow v$$

$$G(V, E)$$

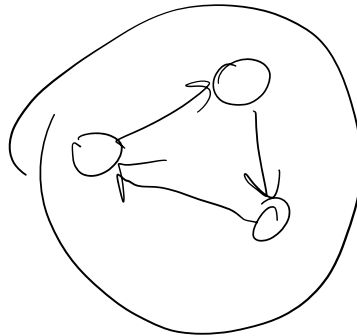
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$$\boxed{(u, v)} \quad u \rightarrow v$$

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- **Path:** A sequence of distinct vertices where each pair of consecutive vertices have an edge

$1, \dots, n$

1 2 3 4 6

Graphs

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 - **Directed:** Each edge $(u, v) \in E$ now has a direction $u \rightarrow v$
 - **Acyclic:** No cycles.
- **Path:** A sequence of distinct vertices where each pair of consecutive vertices have an edge
- **Cycle:** A sequence of distinct vertices where each pair of consecutive vertices have an edge **and** the first and last vertices are connected.

Graph Algorithms: Traversal

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- **BFS:**

- **Purpose:** Reachability, Shortest Path (unweighted graph)
- **Implementation details:** Add your neighbours to a **queue**, pop from the queue to get next node
- **Runtime:** $O(V + E)$

Graph Algorithms: Traversal

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- **DFS:**

- **Purpose:** Reachability, toposort
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Graph Algorithms: Shortest Path

- Dijkstra's

- **Purpose:** SSSP, no negative edges
- **Implementation:** Visit neighbours in **priority queue**
- **Runtime:** $\underbrace{O(m \log n)}$ (with **Fibonacci Heap**, $\underbrace{O(m + n \log n)}$) ✕

Graph Algorithms: Shortest Path

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- **Bellman-Ford:**

- **Purpose:** SSSP, yes negative weights. Will detect negative cycles.
- **Implementation:** Dynamic Programming recurrence:
 - ▶ $d(v, k)$ is the shortest-walk distance from s to v using at most k edges
 - ▶ $d(v, k) = \min \left(d(v, k-1), \min_{u \rightarrow v} d(u, k-1) + \ell(u \rightarrow v) \right)$
- **Runtime:** $O(mn)$

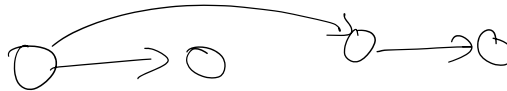
$$m = |E|$$

$$n = |V|$$

Graph Algorithms: Longest Path

Longest path in a Directed Acyclic Graph (DAG)

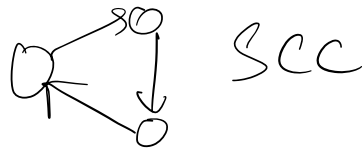
- **Purpose:** To find the longest simple path (no repeating vertices) by weight in a graph which is guaranteed to be a DAG¹.
- **Runtime²:** $O(V + E)$
- **Returns:** The sum of the weights of the longest path in the DAG.



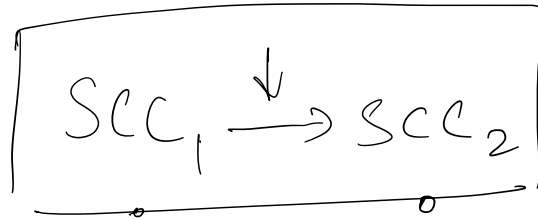
¹Finding the longest path in other types of graphs is at least NP-hard.

²This is a relatively straight-forward DP on a DAG problem if you wish to derive it.

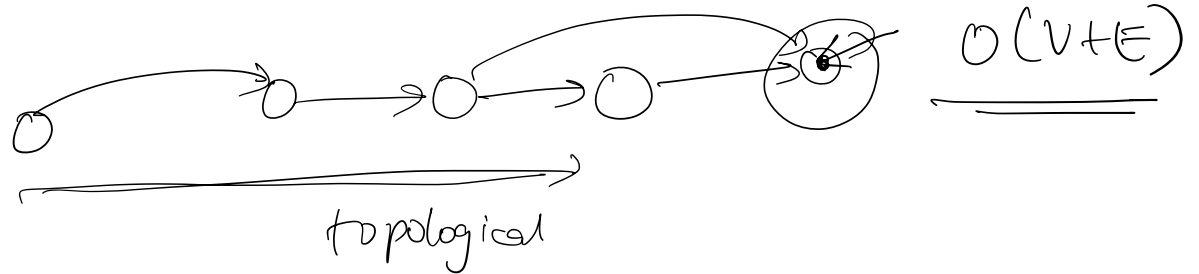
SCC



metagraph $\rightarrow$ DAG



Topo sort



Tarjan's Kosaraju-Sharir

$O(V+E)$

Graph Problems: General Stuff

How to solve graph problems:

1. Identify type of problem (Reachability, Shortest Path, SCC)
2. Construct new graph
 - o Add sources/sinks
 - o Add vertices via $\underline{V'} = \underline{V} \times \{\text{some set}\}$ (Useful for tracking states)
 - o Add vertices via $\underline{E'} = \underline{E} \times \{\text{some set}\}$ (Useful for allowing/prohibit certain behaviour)
3. Apply some stock algorithm (**DO NOT MODIFY THE ALGORITHMS - MODIFY THE INPUTS!**)
4. Draw connection between how to result of the algorithm upon the new graph relates to the solution of the original question.

$$G'(V', E')$$

$$\text{Time complexity} = O(\underset{\uparrow}{n} \mid \underset{\uparrow}{m})$$

$$|V'|$$

Dfs

Recurrences and Asymptotics

Recursion
Tree

Give a tight asymptotic bound for each recurrence: $c_1 374^x \leq 2^x$

- $T(n) = 4 T(\frac{n}{2}) + n \log_2 n$

- $T(n) = T(\frac{3n}{4}) + T(\frac{n}{4}) + 5n$

- $T(n) = 9 T(\frac{n}{3}) + n^2$

$$c_1 g(x) \leq f(x) \leq c_2 g(x)$$

Group the following functions s.t. f and g are in the same group if $f(x) \sim \Theta(g(x))$, and sort the groups by runtime:

- n^n

- ~~$\log \log n$~~

- 374^n

- $n!$

- ~~$\log(n + n^{374})$~~ $\approx \log(n^{374}) = 374 \log n$

- ~~$\log n^n = n \log n$~~

- ~~$n^{1.000001}$~~

- 2^n

- ~~$n \log n^5$~~

- $\frac{1}{\log_n 2} = \frac{1}{\frac{\ln 2}{\ln n}} = \frac{\ln n}{\ln 2} = \log_2 n$

$$\frac{1}{\log_n 2}, \log(n + n^{374})$$

$$\log(n^n), n \log(n^5)$$

$$n^{1.0001}$$

$$2^n$$

$$374^n$$

$$n!$$

$$n^n$$

$$\log n$$

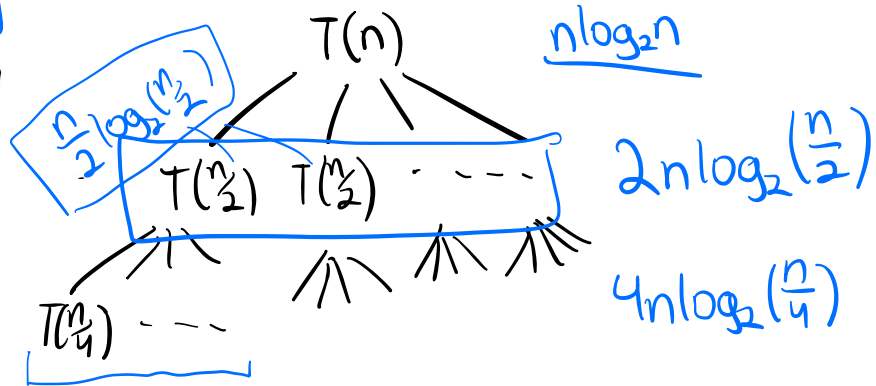
$$n^\varepsilon \gg \log n$$

Recurrences and Asymptotics

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- $T(n) = 9 T(\frac{n}{3}) + n^2$

$$\frac{n}{2^h} = 1 \Rightarrow 2^h = n$$

$$h = \log_2 n$$



$$4^h = 4^{\log_2 n} = n^{\log_2 4} = n^2$$

$$\Downarrow$$

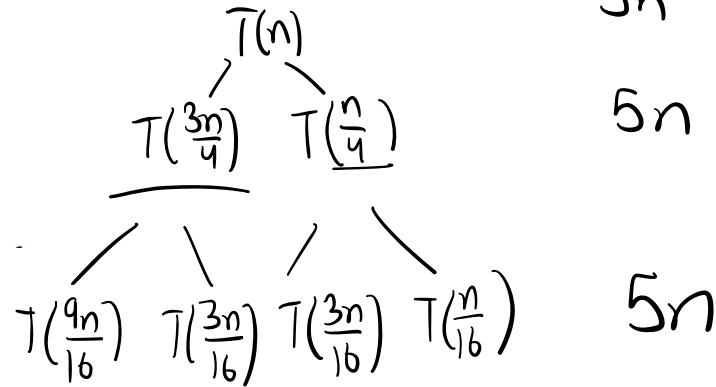
$$2^{2 \log_2 n} = n^2$$

$T(1) \quad T(1) \quad \dots \quad T(1)$

Total: $O(n^2)$

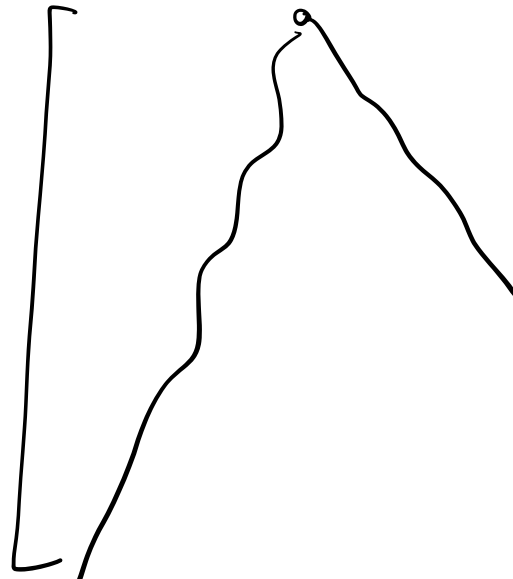
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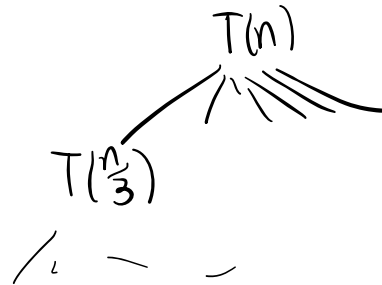
Total: $O(n) \times \# \text{ of levels}$
 $= O(n \log n)$

$\leq \log n$



Recurrences and Asymptotics

- $T(n) = 4 T(\frac{n}{2}) + n \log_2 n$
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- $T(n) = 9 T(\frac{n}{3}) + n^2$



$$\begin{array}{c}
 n^2 \\
 9 \cdot \left(\frac{n}{3}\right)^2 = n^2 \\
 n^2 \\
 \vdots
 \end{array}$$

$$\text{Time: } O(n^2) \times \# \text{ of levels} = \underline{O(n^2 \log n)}$$

Divide and Conquer

Suppose you have two arrays: A of length n_1 , sorted in ascending order, and B of length n_2 , sorted in descending order. You decide to concatenate them to form a new array $C = A \cdot B$ for some application.

All the numbers are distinct. After concatenation, you no longer care about A and B individually, and you don't know n_1 or n_2 . You only have C and its size $n = n_1 + n_2$.

For example, if $A = [1, 2, 6, 7]$ and $B = [8, 4, 3]$, then $C = [1, 2, 6, 7, 8, 4, 3]$.

Given a number x , we want to check whether x appears in C .

Search(C)

if $\text{len}(C) = 1$:
return 1

$n = \text{len}(C)$

$m = \frac{n}{2}$

$d = C[m]$

$l = C[m-1]$

$r = C[m+1]$

if $l \leq d \leq r$

return $\text{Search}(C[m+1:n]) + m$

else if $r \leq d \leq l$

return $\text{Search}(C[0:m-1])$

o/w
return m

Answer(C, x)

$s = \text{Search}(C)$

run Binary
Search on

$C[1..s]$ and

$C[s+1..n]$

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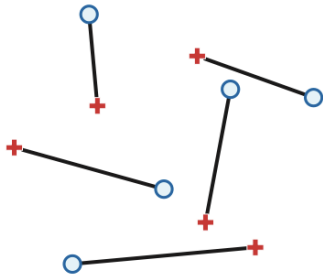
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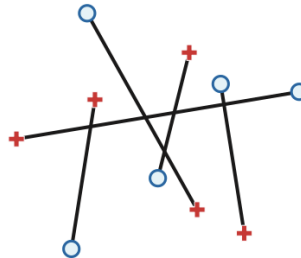
Divide and Conquer

Given sets $R, B \subset \mathbb{R}^2$ (n red points and n blue points) in the plane, we define

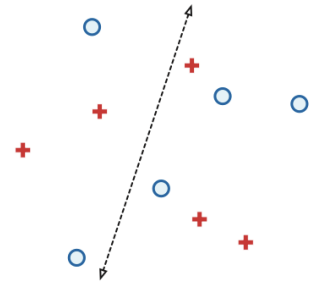
- **Tidy Matching:** n non-intersecting segments pairing each red point to a unique blue point. (Fig. 1(a) vs 1(b))
- **Ham-sandwich cut ℓ :** A line with exactly $\lfloor n/2 \rfloor$ red and $\lfloor n/2 \rfloor$ blue points strictly above it. (Fig. 1(c))



(a) Tidy Matching



(b) Not a tidy matching



(c) Ham-Sandwich Cut

Fact: A valid tidy matching and ham-sandwich cut always exist.

Black-Box Subroutines:

- **HamSammy** (R, B): Returns a ham-sandwich cut ℓ in $O(n)$ time.
- **Above?** (p, ℓ): Returns **True** if p is strictly above ℓ in $O(1)$ time.

Describe and analyze an algorithm to compute a Tidy Matching of R and B , using these subroutines.

Divide and Conquer

TM = self()

TidyMatching(R, B):

$O(n \log n)$

$n = |R|$

if $n = 0$: return

if $n = 1$:

TM.add(R[0], B[0])

$T(n) = 2T(n/2) + O(n)$

$O(n) \leftarrow \text{work/level}$
 $\log n \approx \text{no. of levels}$

$l = \text{HAMMIDITY}(R, B) \leftarrow O(n) \text{ time}$

$O(n)$ { $R_UP = \text{set of all points in } R \text{ s.t. ABOVE}(p, l) \text{ is true}$
 R_DOWN
 B_UP, B_DOWN $n/2$ $n/2$
TightMatching(R_UP, B_UP); TightMatching(R_DOWN, B_DOWN)

TidyMatch(R, B):

if $n = 0$:
return

if $n = 1$:
return $(R[0], B[0])$

R_UP =

R_DOWN =

B_UP =

B_DOWN =

} use ABOVE(p, l)

return TidyMatch(R_UP, B_UP) $\cup$ TidyMatch
(R_DOWN, B_DOWN)

Dynamic Programming

A *shuffle* of two strings X and Y is formed by interspersing the characters into a new string, keeping the characters of X and Y in the same order. For example, the string **BANANAANANAS** is a shuffle of the strings **BANANA** and **ANANAS** in several different ways.

BANANAANANAS **BANANAANANAS** **BANANAANANAS**

Similarly, the strings **PRODGYRNAMAMMIINCG** and **DYPRONGARMAMMICING** are both shuffles of the strings **DYNAMIC** and **PROGRAMMING**:

PRODGYRNAMAMMIINCG | **DYPRONGARMAMMICING**

Given three strings $A[1..m]$, $B[1..n]$, and $C[1..m+n]$, describe and analyze an algorithm to determine whether C is a shuffle of A and B .

DP

Let $\text{Shuffle}(i, j)$ be the true/false value that represents whether $C[1 \dots i+j]$ is a shuffle of $A[1 \dots i]$ and $B[1 \dots j]$

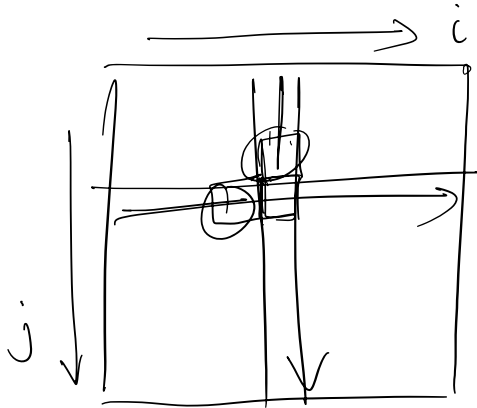
$$\text{Shuffle}(i, j) = \begin{cases} \text{True} & \text{if } j=0 \text{ \& } i=0 \\ (i > 0 \wedge \underline{A[i] = C[i+j]} \wedge \text{Shuffle}(i-1, j)) \vee (j > 0 \wedge \underline{B[j] = C[i+j]} \wedge \text{Shuffle}(i, j-1)) & \text{o/w} \end{cases}$$

Qs Array Shuffle DP $[0 \dots m, 0 \dots n]$

Eval order : i in incr , j in ince

Soln Loc : Shuffle DP $[m, n]$

RunTime : $O(mn)$



$O(mn)$

for i in $[1 \dots m]$
for j in $[1 \dots n]$

What *is* it about cephalopods?

An entrepreneurial octopus runs a "Fish Wash" at the local coral reef. At exactly 8:00 AM, a morning rush of n fish arrive to be cleaned.

To handle the rush, the octopus acts as a multi-lane car wash, dedicating exactly **3 of its arms** to scrub the fish.

Each fish i (for $1 \leq i \leq n$) requires ℓ_i minutes to scrub clean, where ℓ_i is a positive integer at most L . The octopus assigns each fish to one of the 3 arms. Fish assigned to the same arm form a line and must be cleaned in their original sequential order.

Describe an algorithm to assign the fish to the 3 arms in a way that **minimizes the sum of the waiting times** for all the fish. Analyze the runtime of your algorithm in terms of n and L .

Example: $\ell = [5, 3, 10, 20, 6, 15]$ and $L = 20$. Suppose the octopus assigns fish 1 and 3 to Arm 1; fish 2, 4, and 5 to Arm 2; and fish 6 to Arm 3.

- Fish 3 waits 5 mins (while Arm 1 cleans Fish 1).
- Fish 4 waits 3 mins (while Arm 2 cleans Fish 2).
- Fish 5 waits $3 + 20 = 23$ mins (while Arm 2 cleans Fish 2 and 4).

The total waiting time is $5 + 3 + 23 = 31$ minutes.

DP

Dynamic Programming

The **StupidScript** language includes a binary operator $@$ that computes the *average* of its two arguments. For example, the StupidScript code `print(3 @ 6)` would print **4.5**, because $(3 + 6)/2 = 4.5$.

Expressions like $3@7@4$ that use the $@$ operator more than once yield different results when they are evaluated in different orders:

$$(3@7)@4 = 5@4 = 4.5 \quad \text{but} \quad 3@(7@4) = 3@5.5 = 4.25$$

Here is a larger example:

$$\begin{aligned} (((((8@6)@7)@5)@3)@(0@9)) &= 4.5 \\ ((8@6)@(7@5))@((3@0)@9) &= 5.875 \\ (8@(6@(7@(5@(3@0))))))@9 &= 7.890625 \end{aligned}$$

Your goal for this problem is to describe and analyze an algorithm to compute, given a sequence of integers separated by $@$ signs, the **largest possible** value the expression can take by adding parentheses. Your input is an array $A[1 \dots n]$ listing the sequence of integers.

For example, if your input sequence is $[3, 7, 4]$, your algorithm should return 4.5, and if your input sequence is $[8, 6, 7, 5, 3, 0, 9]$, your algorithm should return 7.890625. Assume all arithmetic operations (including $@$) can be performed exactly in $O(1)$ time.

DP

Bad Apple

Jeff is falling into a Bad Apple rabbit hole on Youtube. He wants to get from the original **Bad Apple** video to the video **bad apple but it's a sorting algorithm**³ by *only* watching videos that are recommended to him in sequence.

Formally, you are given a list of n videos V such that each $v \in V$ has a watch time

$\tau(v)$ and list $R(v) \subseteq V$ of sidebar recommended videos that can be watched after v is watched. Suppose video $s \in V$ is the original **Bad Apple** video and $t \in V$ is the **bad apple but it's a sorting algorithm** video. We call a sequence of videos starting from s and ending at t a Terpsichorean marathon.

- (a) Describe and analyze an algorithm to find a Terpsichorean marathon of minimum total watch time.
- (b) We call a video a *banger* if it is in **some** Terpsichorean marathon with minimum watch time. Design and analyze an efficient algorithm to find all the *banger* videos.

Make a graph $G = (V, E)$, where $E = \{(u, v) \mid v \in R(u)\}$

Then any (s, t) path in G is a marathon.

a) Set $\ell(u, v) = \tau(u)$. Then we need the shortest (s, t) path.

³Yes, this is a real thing.

Dijkstra's returns $d(s,u)$ for all $u \in V$.

Graphs

Run Dijkstra's from s ; return shortest (s,t) path.

Runtime: $O(|E| \log |V|) = O(m \log n)$, where $m = \sum_{v \in V} |R(v)|$.

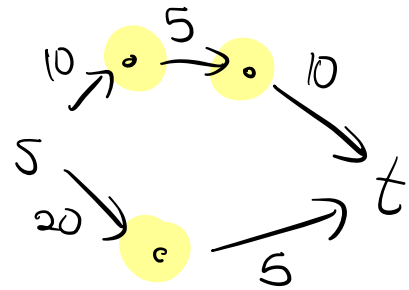
b) u is a banger iff

$$\underline{d(s,u)} + \underline{d(u,t)} = d(s,t)$$

Compute $d(s, \cdot)$ w/ Dijkstra's from s .

Compute $d(\cdot, t)$ w/ Dijkstra's from t in reverse graph.

Return all $u \in V$ where $d(s,u) + d(u,t) = d(s,t)$.



Runtime: $O(m \log n + m \log n + n) = \underline{O(m \log n)}$.

Who's the number one spice?

Young Cardamom is bringing the flavor to the fish!⁴ His dish needs to have both **HEAT** and **SWEET** spices to be complete.

The kitchen can be modeled as a DAG where:

- Vertices represent prep stations
- Each station is labeled either HEAT, SWEET, or NEUTRAL
- A directed edge (u, v) with weight $\cancel{f(u, v)}$ means station u can send ingredients to station v , contributing flavor value $f(u, v)$
- Young Cardamom starts at station s and must reach plating station t

To achieve the #1 spice, Young Cardamom's path must visit at least one HEAT station and at least one SWEET station.

Design and analyze an algorithm to find a path from s to t that visits both required station types and maximizes the total flavor value.

Idea #1: New graph $G \times \{0, 1\} \times \{0, 1\} \Rightarrow$ long-path from $(s, 0, 0)$ to $(t, 1, 1)$.

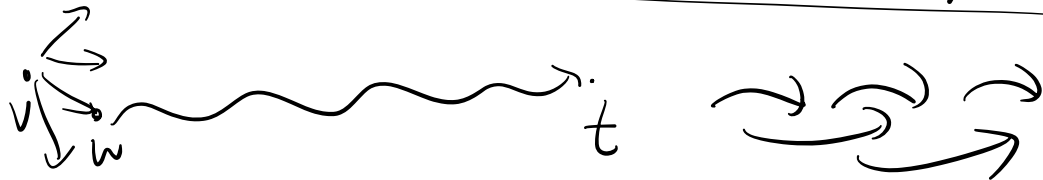
⁴Of course, HAB is bringing the flavor to the rice.

Graphs

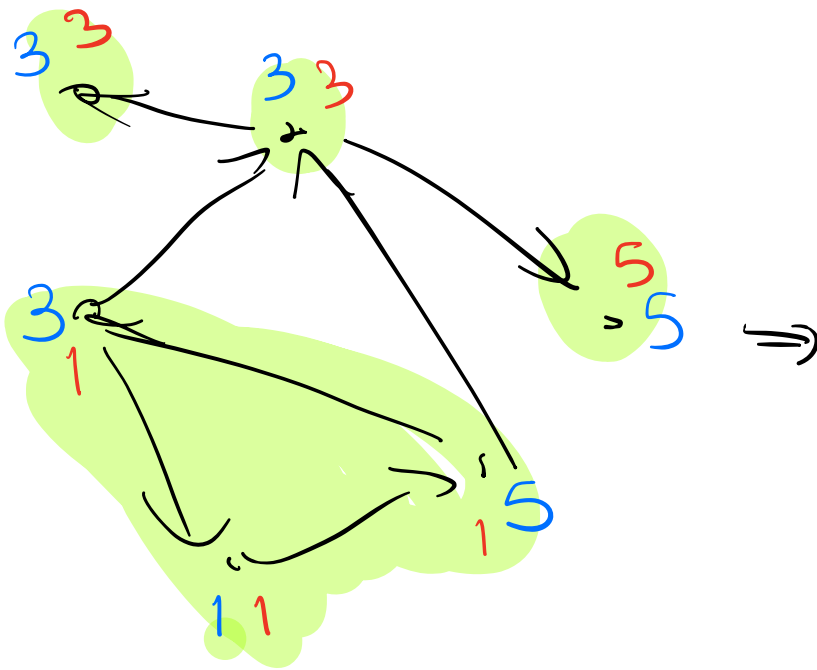
$\text{BestPath}(v, \underline{a}, \underline{b}) = \text{length of LP from } v \text{ to } t \text{ where}$
 we visit at least a sweet & b heat.

Want: $\text{BestPath}(s, 1 - [s \text{ is sweet}], 1 - [s \text{ is spicy}])$

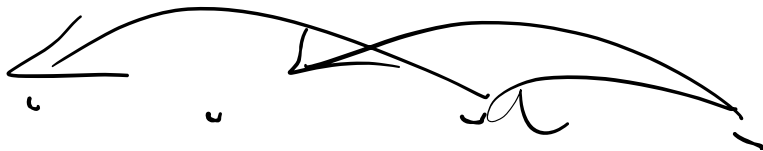
$$\text{BestPath}(v, a, b) = \begin{cases} 0 & \text{if } v=t, a=b=0 \\ -\infty & \text{if } v=t, \text{ } \\ \max_{(v,w) \in E} \left\{ f(v,w) + \text{BestPath}(w, \right. \\ \quad \left. \max(0, a - [w \text{ is sweet}]) \right. \\ \quad \left. \max(0, b - [w \text{ is heat}]) \right\} \end{cases}$$

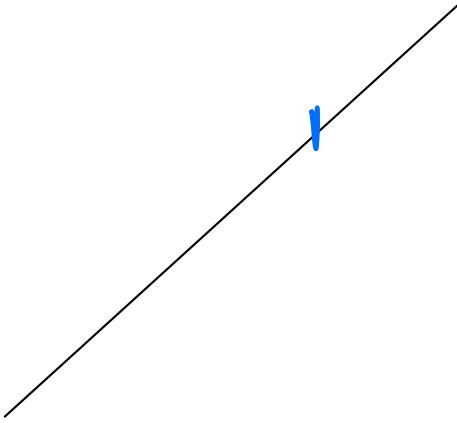


Eval order: Reverse topo for v inc a & b $\leftarrow \leftarrow \leftarrow \leftarrow$

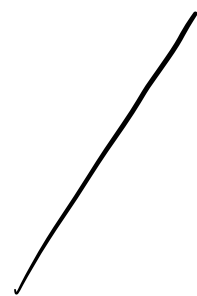


$$\text{MinW}(c) = \min \left\{ w(c), \min_{\substack{(c,d) \in E^{\text{sc}}}} \text{MinW}(d) \right\}$$





$\Rightarrow$



1 2 3 4 5

4 5 12 3

$A[1 \dots n]$

GetMax(i, j)

if $j - i < 374$
brute force

$m \leftarrow \left\lfloor \frac{i+j}{2} \right\rfloor$

if $A[m] > A[0]$ (m is left half)

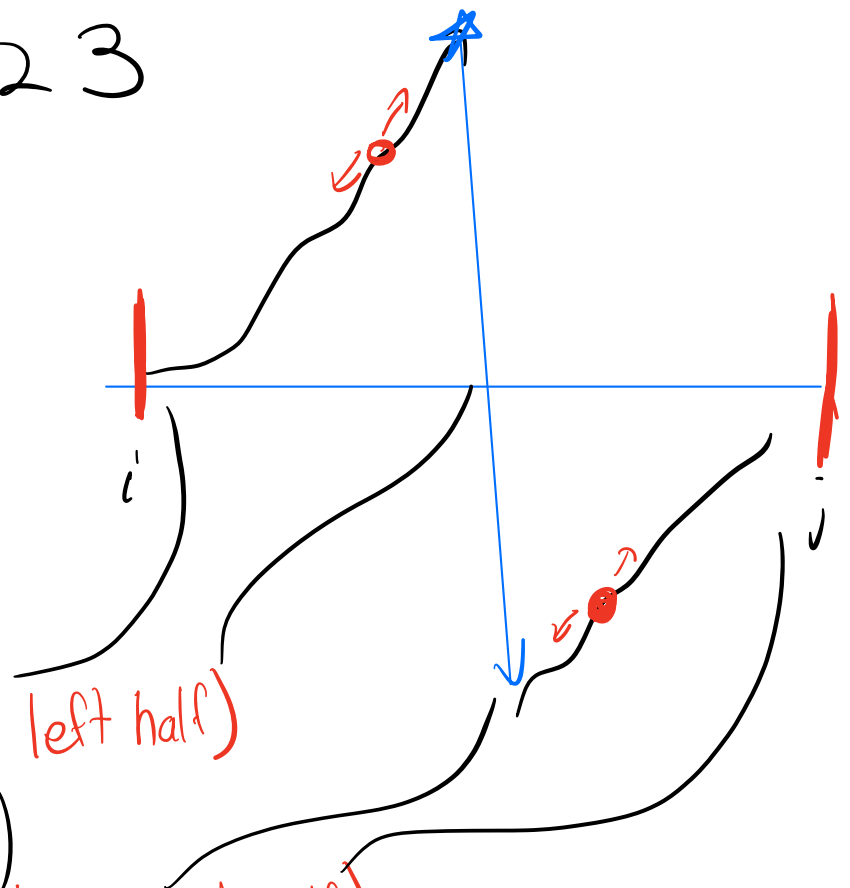
return GetMax(m, j)

if $A[m] < A[0]$

return GetMax(i, m) (m is right half)

if $A[m] = A[0]$

die



Runtime: $O(m+n)$.

Graphs

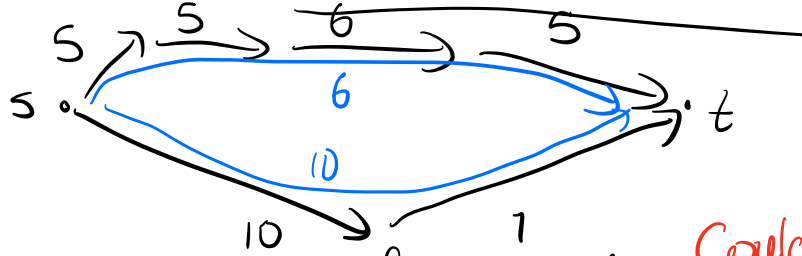
We are given a directed graph $G = (V, E)$ with n vertices and m edges, where $m \geq n$. Each edge $e \in E$ has a weight $w(e) \geq 0$, and all weights are distinct. Let s and t be two distinct vertices in V .

For a path π from s to t , we define its **bottleneck value** as

$$b(\pi) = \max_{e \in \pi} w(e),$$

i.e., the weight of the heaviest edge along π .

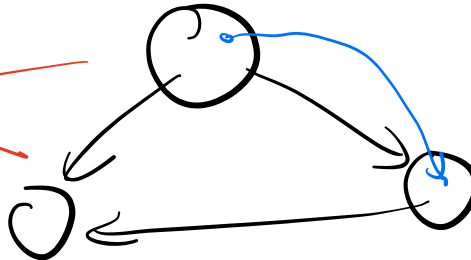
We want to find a path from s to t that **minimizes the bottleneck value**.



~~DP? $\text{MinB}(u) := \min \text{bottleneck from } u \rightarrow t$.~~

Could have cycles!

~~DP on SCC graph?~~



Graphs

Modify Dijkstra's: Will work, but should avoid modifying algo's.

Q: Can we determine if bottleneck is $\leq L$?

A: Remove all e s.t. $w(e) > L$ & run BFS.

Takes $O(n + m)$ time.

Can use this w/ bin. search!

$O(m \log n)$

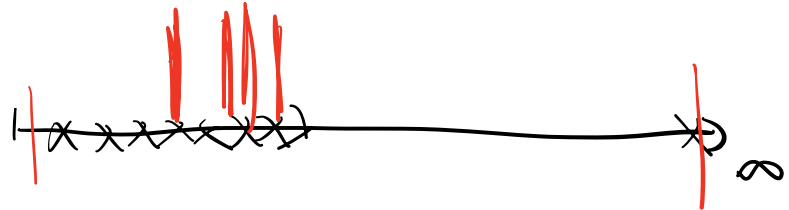
Sort edges increasing.

$l \leftarrow 0$

$u \leftarrow n$

$M \leftarrow$ weight of edge $\frac{l+u}{2}$

$l \leftarrow \frac{l+u}{2}$



else
u_k $\frac{L+u}{2}$

Feedback

- Further questions in ACM (Siebel 1104) after this
- Please fill out the feedback form:
`go.acm.illinois.edu/cs374a_mt2_feedback`

